

Marcus Greenan

La Jolla, CA

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Mechanical Engineering student specializing in robotics and controls with hands-on aerospace structures, autonomous systems, CAD/FEA, machining, and experimental validation experience. Strong foundation in dynamics, feedback control, mechanical design, system integration, hardware testing, and rapid design iteration for aerospace and electromechanical systems.

EDUCATION

University of California, San Diego (UCSD)

La Jolla, CA

B.S. Mechanical Engineering, Specialization in Robotics and Controls

Provost Honors – GPA: 3.61 – Expected Graduation: June 2027

Relevant Coursework: Dynamics, Control Systems, Fluid Mechanics, Solid Mechanics, Thermodynamics

TECHNICAL SKILLS

Aerospace & Structures: CAD/FEA, Structural Design, Aerodynamic Trade Studies, OpenRocket

Robotics & Controls: ROS, ROS2, PID Tuning, GPS Path Tracking, System Identification

Programming: MATLAB, Python, C++, Linux, Data Analysis, Automation Scripts

Mechanical Design: SolidWorks, Fusion 360, AutoCAD, CNC Machining, GD&T, DFM

Testing: Experimental Validation, Data Logging, Hardware Integration, Root Cause Analysis

PROJECTS

Rocket Propulsion Lab – Daedalus

10/2024 – 6/2025

Structures Lead

- Designed and validated structural rocket components using CAD modeling and FEA.
- Conducted aerodynamic trade studies supporting stability and performance validation.
- Collaborated with propulsion and avionics teams to integrate flight subsystems.
- Supported build, assembly, and test work for hands-on rocket hardware development.
- Evaluated structural tradeoffs involving mass, stiffness, manufacturability, and integration.

MAE 148 – Autonomous Trash Can Retrieval Robot

1/2026 – Present

Mechanical Lead

- Led mechanical subsystem design within six-member interdisciplinary robotics team.
- Designed and fabricated custom 5-axis robotic arm with actuator integration.
- Integrated drivetrain, steering, and control architecture with ROS navigation stack.
- Optimized path tracking and system performance through iterative field validation.
- Built and modified hardware to improve reliability during autonomous outdoor testing.

UCSD Gravish Lab

4/2026 – Present

Research Assistant

- Design and fabricate soft robotic systems for bio-inspired robotics and manipulation.
- Develop mechanical prototypes and test fixtures using CAD, machining, and prototyping.
- Program robotic hardware using Python and C++ for testing and data collection.
- Work with PhD mentors to iterate hardware designs based on experimental results.

Multi-Agent Robotics Coverage

10/2025 – Present

Undergraduate Researcher

- Developed Voronoi-based coverage and centroid control algorithms for TurtleBot systems.
- Integrated localization, mapping, and ROS feedback control for distributed operation.
- Tuned controller gains using Python logging and experimental system identification.
- Validated convergence, coverage efficiency, and robustness through testing.

EXPERIENCE

UCSD Jacobs Machine Shop

1/2026 – Present

Tutor / Assistant

- Machined aluminum and polymer components maintaining tight dimensional tolerances.
- Interpreted engineering drawings and GD&T for fabrication and inspection.
- Assisted teams in fabrication planning, fixturing design, and dimensional inspection.
- Supported design-for-manufacturing decisions for robotics and mechanical projects.

iLAB BioTech Partners

6/2022 – 9/2023

Software Engineering Intern

- Processed large-scale datasets using Python with structured data pipelines.
- Applied statistical analysis and automation workflows to support technical research.

Joint BioEnergy Institute (JBEI)

6/2022 – 9/2022

Engineering Intern

- Conducted experimental validation and data analysis for controlled biological systems.
- Presented technical findings to researchers, mentors, and peers.

LEADERSHIP EXPERIENCE

Tau Kappa Epsilon – UCSD Chapter

1/2025 – 1/2026

Treasurer

- Managed \$75K+ operating budget including allocation planning and reporting.
- Implemented structured tracking improving transparency and long-term forecasting.